

Compact-Group Quotients Preserve Hilbert Representability and an Affirmative Answer to Problem 8.1(c)

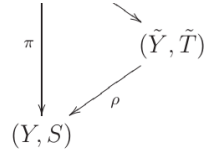
Candidate full solution – run `fa.banach.001`, lane 14

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Candidate full answer to option (c); likely valid. Every metrizable recurrent-transitive Hilbert system is an almost one-to-one factor of a Hilbert-representable system. Options (a), (b), and Problem 8.2 are not claimed.

Source question

Glasner and Weiss [1] prove in their Theorem 3.1 that every metrizable recurrent-transitive Hilbert system has an almost one-to-one extension which is a compact group-factor of a Hilbert-representable system. On source PDF page 13, Problem 8.1 asks whether either extension can be removed and lists three precise alternatives.



with $Z \subset X$ a subsystem, σ a group-extension and ρ an almost 1-1 extension.

Proof. Just repeat the steps 2. to 5. of the proof of Theorem 3.1 above. $\square$

8. SOME OPEN PROBLEMS

8.1. Problem. In Theorem 3.1 we have shown that every metrizable recurrent-transitive Hilbert system admits an almost 1-1 extension which is a group-factor of a Hilbert-representable system. Can one get rid in this structure theorem of either one of these extensions or maybe of both? Thus, our question is whether a metrizable recurrent-transitive Hilbert system is always: (a) Hilbert-representable, or (b) a group-factor of a Hilbert-representable system, or (c) an almost 1-1 factor of a Hilbert-representable system. Question (a) can be reformulated as follows: Is a factor of a Hilbert-representable system also Hilbert-representable? This latter question is stated as Problem 996 in [21].

8.2. Problem. Is there a metrizable recurrent-transitive Hilbert system (X, T) with no nontrivial Hilbert-representable factors? In other words, is there a nontrivial $\mathbb{Z}$ -algebra $\mathcal{A} \subset H(\mathbb{Z})$ with $\mathcal{A} \cap B(\mathbb{Z}) = \mathbb{C}$? Such a system will provide a counter-example to option (a) in Problem 8.1.

Figure 1: The structure-theorem context and Problems 8.1–8.2 in [1], source PDF page 13.

Selected transcription (Problem 8.1(c)). Is a metrizable recurrent-transitive Hilbert system always “an almost 1-1 factor of a Hilbert-representable system”?

Definitions and source input

A compact metric dynamical system (X, T) is *Hilbert-representable* if there are a Hilbert space $\mathcal{H}$, a unitary operator U on $\mathcal{H}$, and a weakly compact U -invariant set $Z \subset \mathcal{H}$ such that (X, T) is isomorphic to $(Z, U|_Z)$. A factor map is *almost one-to-one* if the points in the target having singleton fibers form a dense set (equivalently in this compact metric setting, a dense G_δ set).

The only dynamical input used below is the construction in the proof of [1, Theorem 3.1]. It gives a Hilbert-representable point-transitive system (X, U) , a compact subgroup K of its uniform Ellis group Λ_X , and a diagram

$$X \longrightarrow \tilde{Y} = X/K \xrightarrow{\rho} Y,$$

where the first map is the K -orbit quotient and ρ is almost one-to-one.

Proof intuition

The apparent loss of Hilbert representability occurs at the quotient X/K . To recover it, replace a point x by the probability measure uniformly distributed on its compact orbit Kx . Encode this measure by all of its symmetric tensor moments. The moments live in Hilbert tensor powers and transform unitarily under the dynamics. They separate orbit measures because polynomials in weak linear coordinates are dense in the continuous functions on the weakly compact set. A weighted direct sum of all moments therefore embeds X/K into one weakly compact Hilbert system.

The compact group in the source is initially a group of homeomorphisms of X . A short cyclicity argument lifts it to strongly continuous commuting unitaries, so the moment construction applies to exactly the quotient in Theorem 3.1.

Compact-group quotient lemma

Write $\mathcal{H}_{\mathbb{R}}$ for the realification of a real or complex Hilbert space, and $\mathcal{H}_{\mathbb{R}}^{\odot n}$ for its n th real symmetric Hilbert tensor power.

Lemma 1 (Tensor-moment quotient). *Let $X \subset \mathcal{H}$ be weakly compact and invariant under a unitary U . Let K be a compact group with a strongly continuous unitary representation on $\mathcal{H}$ such that every $k \in K$ preserves X and commutes with U . Then the quotient dynamical system $(X/K, \bar{U})$ is Hilbert-representable.*

Proof. Pass to $\mathcal{H}_{\mathbb{R}}$, on which all the operators are orthogonal and the weak topology on X is unchanged. Let m_K be normalized Haar measure. For $n \geq 1$ define

$$M_n(x) = \int_K (kx)^{\otimes n} dm_K(k) \in \mathcal{H}_{\mathbb{R}}^{\odot n}.$$

This Bochner integral exists because the orbit map is norm-continuous. It is constant on K -orbits by invariance of Haar measure.

Each $M_n : X \rightarrow \mathcal{H}_{\mathbb{R}}^{\odot n}$ is weak-to-weak continuous. Indeed, for a pure symmetric tensor $v_1 \odot \cdots \odot v_n$, its scalar pairing with $M_n(x)$ is an integral of

$$\prod_{j=1}^n \langle kx, v_j \rangle = \prod_{j=1}^n \langle x, k^{-1}v_j \rangle.$$

This is jointly continuous in $(k, x) \in K \times X$: the vectors $k^{-1}v_j$ vary in norm and X is bounded in norm. Integration over compact K preserves continuity. Finite linear combinations of pure tensors are

dense, and the estimate $\|M_n(x)\| \leq R^n$, where $R = \sup_{x \in X} \|x\|$, extends the conclusion to every test tensor.

The moments separate K -orbits. To see this, let μ_x be the pushforward of m_K under $k \mapsto kx$. If $M_n(x) = M_n(y)$ for all $n \geq 1$, then μ_x and μ_y have the same integrals against every finite product of real weak linear coordinates. The algebra of such real polynomials contains the constants and separates points of the weakly compact Hausdorff space X . The real Stone–Weierstrass theorem [3] gives $\mu_x = \mu_y$. Haar measure has full support, so

$$\text{supp } \mu_x = Kx, \quad \text{supp } \mu_y = Ky.$$

Thus $Kx = Ky$.

Choose positive weights $a_n = 2^{-n}(1+R)^{-n}$ and set

$$\Phi(x) = \bigoplus_{n=1}^{\infty} a_n M_n(x) \in \mathcal{K} := \bigoplus_{n=1}^{\infty} \mathcal{H}_{\mathbb{R}}^{\odot n}.$$

The bound $a_n \|M_n(x)\| \leq 2^{-n}$ gives a uniformly square-summable tail, so coordinatewise weak continuity implies weak continuity of Φ . The map descends to a continuous injection $\bar{\Phi} : X/K \rightarrow \mathcal{K}$. Since X/K is compact and the weak topology is Hausdorff, this is a homeomorphism onto a weakly compact set.

Finally, commutation of U with K gives

$$M_n(Ux) = U^{\odot n} M_n(x).$$

Hence $\bar{\Phi}$ intertwines $\bar{U}$ with the orthogonal operator $\bigoplus_{n \geq 1} U^{\odot n}$ on $\mathcal{K}$. This is a Hilbert representation of the quotient system. $\square$

Lifting the compact kernel

Lemma 2 (Uniform Ellis group lift). *Let (X, U) be a point-transitive Hilbert-representable metric system. After choosing a cyclic Hilbert realization*

$$X = \text{weakcl}\{U^n x_0 : n \in \mathbb{Z}\}, \quad \mathcal{H} = \overline{\text{span}}\{U^n x_0 : n \in \mathbb{Z}\},$$

the uniform Ellis group Λ_X has a strongly continuous unitary representation on $\mathcal{H}$. It extends its action on X , and its image commutes with U .

Proof. Every transitive point $z \in X$ has $\|z\| = \|x_0\|$. Indeed, as a weak limit of orbit points it satisfies $\|z\| \leq \|x_0\|$. Since x_0 is in the weak orbit closure of the transitive point z , weak lower semicontinuity gives the reverse inequality.

For $g \in \Lambda_X$, choose integers n_j for which $U^{n_j}|_X$ converges uniformly to g in a compatible weak metric. Then $U^{n_j} x_0 \rightarrow g x_0$ weakly. The point $g x_0$ is transitive, so the preceding norm equality upgrades this convergence to norm convergence. On every vector $p(U)x_0$ in the algebraic cyclic span,

$$U^{n_j} p(U)x_0 = p(U)U^{n_j} x_0$$

therefore converges in norm. Uniform boundedness and density give a strong operator limit $\hat{g}$ on $\mathcal{H}$. Applying the same construction to g^{-1} shows that $\hat{g}$ is unitary. It is independent of the approximating sequence, extends g on X , and commutes with U .

If $g_i \rightarrow g$ in Λ_X , then $g_i x_0 \rightarrow g x_0$ weakly; all these points are transitive and have the same norm. Thus the convergence is in norm, and cyclicity gives $\hat{g}_i \rightarrow \hat{g}$ strongly. The lift is strongly continuous. $\square$

Result

Theorem 1 (Affirmative answer to Problem 8.1(c)). *Every metrizable recurrent-transitive Hilbert system is an almost one-to-one factor of a Hilbert-representable system.*

Proof. Let (Y, T) be such a system. Apply the construction of [1, Theorem 3.1]. It supplies a point-transitive Hilbert-representable system (X, U) , a compact subgroup $K \leq \Lambda_X$, the quotient $\tilde{Y} = X/K$, and an almost one-to-one factor map $\rho : \tilde{Y} \rightarrow Y$.

By the uniform Ellis group lift, K acts strongly continuously on a cyclic Hilbert realization of X by unitaries commuting with U . The tensor-moment quotient lemma therefore shows that $\tilde{Y} = X/K$ is Hilbert-representable. The already constructed map $\rho : \tilde{Y} \rightarrow Y$ is almost one-to-one, which is exactly option (c). $\square$

Verification, limitations, and novelty

The proof uses no selection theorem and no computational assumption. The moment map is defined on the whole quotient, not merely on a dense orbit. Its separation argument uses all tensor degrees, so it does not assume finite generation of compact-group invariants.

The conclusion does not establish that Y itself is Hilbert-representable (option (a)), that Y is a compact group-factor of a Hilbert-representable system (option (b)), or the algebraic assertion in Problem 8.2.

Bounded searches on 9 August 2026 used the exact source question and arXiv id, “factor of a Hilbert-representable system”, “almost 1-1” with Hilbert representability, compact-group quotients, and symmetric tensor moments. They found later surveys repeating the question. Ben Yaacov–Ibarlucía–Tsankov [2] prove factor closure for the Hilbert compactification of a special pro-oligomorphic class, not the general quotient lemma or option (c) above. No later general answer was found. This is a bounded search, not a definitive priority claim.

Human-review recommendation

Review first the lift of Λ_X to the strong unitary closure, especially the weak-plus-equal-norm step on the cyclic transitive vector. Then check that equality of the averaged tensor moments determines the Haar orbit measure via real Stone–Weierstrass. These are the only non-source ingredients carrying the conclusion.

References

- [1] E. Glasner and B. Weiss, *On Hilbert dynamical systems*, arXiv:1005.0230, 2010.
- [2] I. Ben Yaacov, T. Ibarlucía, and T. Tsankov, *Eberlein oligomorphic groups*, Trans. Amer. Math. Soc. 370 (2018), 2181–2209; arXiv:1602.05097.
- [3] W. Rudin, *Real and Complex Analysis*, 3rd ed., McGraw–Hill, 1987.